

Enhancing AI Education Through Educational Games: Evidence From a Meta-Analysis Study

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Abstract—The purpose of this paper is to examine the effects of educational games in AI education through a meta-analysis of 22 empirical studies published over the last decade. We examined the effects of different school segment, experimental period, experimental scales, game types, knowledge types, and teacher interventions on students' knowledge achievement, motivation, and thinking skills. The results of the study proved that educational games have an overall positive impact on AI education. Moderator analyses indicated that the games have a greater impact on students' motivation and knowledge achievement, and a lesser impact on thinking skills. Unexpectedly, the longer the experimental period, the smaller the experimental period, and the program with teacher intervention, the better the teaching and learning outcomes. In terms of game types, puzzle and robotics games were the most effective. And on the type of AI knowledge, the study found that games were most effective when used as content to teach AI algorithms.

Keywords—educational game, AI education, meta-analysis

I. INTRODUCTION

In recent years, the rapid development of AI technology is having a huge impact on people's productive lives. AI is also now seen by many as an indispensable driver of the fourth industrial revolution, which may trigger the fourth education revolution[1]. In this context, it is important for students to develop a common understanding of AI and acquire basic knowledge and skills [2]. Many countries have started to deploy strategic plans for next-generation talent training to build a first-mover advantage in national AI development [3]. AI education is starting to become part of the school curriculum.

However, there is still no academic consensus on how exactly AI education should be conducted and how the evaluation system should be constructed. Some studies have shown that the representative content of AI education is based on esoteric and difficult algorithms and abstract concepts, which is difficult for novice learners to understand and master, leading to the inefficiency of AI courses [4]. Moreover, AI courses do not have a unified content framework and knowledge system so far, which makes their teaching more complex and difficult to

learn [5]. Therefore, the most important thing to consider in AI education is the teaching method.

Some scholars have suggested the use of educational games in the learning process of AI education[6]. The rapid development of internet technology and the popularity of devices such as tablets, whiteboards and laptops have made it possible to use educational games in the teaching and learning process. More and more teachers are interested in using educational games as a teaching tool to help students learn. Some empirical studies have shown that educational games are effective in increasing students' motivation and knowledge of AI algorithms[7]. However, there are also studies that take the opposite view, arguing that the motivation for learning triggered by educational games is not intrinsic and can lead to learner distraction and sometimes even negatively affect academic performance[8][9]. Therefore, it is not known whether educational games are actually beneficial for AI education, and this study sees the need for a systematic literature review on this issue.

Previous studies have discussed the use of educational games in AI education, but their effectiveness has not been fully validated. For example, Petri summarized the application of game-based instructional models in computer education by analyzing the research design, evaluation models and analysis tools used in the experiments[10]. Thielscher conducted a cluster analysis of game types in AI education, focusing on the different applications of game types to different AI education topics and content[11]. On the other hand, Zhan and colleagues systematically reviewed the specific factors and interventions of educational games in AI education, with a particular focus on the effectiveness of game-based interventions[12]. However, the available reviews are systematic literature reviews of the use of educational games in AI education, which introduces potential bias and subjectivity into the discussion of educational effectiveness. Although there have been studies conducted in the areas of game-based instruction, educational games, etc., such as Michaela and others who used meta-analysis to gain insight into whether or not games would have an impact on learning in STEM courses for all student populations[13]. However, there is a lack of research in this area in the field of AI education.

Therefore, at this stage, it is necessary to conduct a systematic meta-analysis of the educational impact of

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educational games in AI education, to test the efficiency of educational games in AI education through a series of subgroup analyses, and to specifically analyses which factors influence the educational impact of educational games in AI education through moderating variables. The purpose of this study is to examine the impact of educational games in AI education through a meta-analysis of empirical studies published over the past decade, to provide a basis for relevant theory building, and to provide educators with an up-to-date understanding of the field.

II. PURPOSE OF THIS STUDY

In this study, 46 effect sizes were extracted from 22 studies published between 2010 and 2023 to provide a literature review of gaming interventions in AI education. The presence of publication bias, heterogeneity, overall validity, subgroup validity and between-group variance in this literature is explored in the text to provide a relatively complementary and exhaustive picture for future related research. In particular, we have examined the effect of educational games on the impact of an AI curriculum across a range of school segment, learning outcomes, experimental period, experimental scale, knowledge type, game type and teacher interventions. The meta-analysis aims to answer the following research questions: Q1: What is the overall effect of educational games on AI education? Q2: What is the impact of educational games on students' knowledge achievement, motivation and thinking skills? Q3: Is there a difference in the effectiveness of educational game applications between K-12 and higher education? Q4: Do different experimental period have different effects on gamified AI education? Q5: Do different experimental scale have different effects on gamified AI education? Q6: Do educational games have different effects on different types of AI knowledge? Q7: Do different types of games have different effects on AI education? Q8: Would teacher intervention be more effective for gamified AI education?

III. METHOD

A. Literature Search

In this study, the literature collection was divided into two phases. The first phase was mainly an initial literature search with keywords from three databases, web of science, Scopus and Google scholar, with a deadline of 18 August 2023 for the search. The second phase was a snowballing method to ensure the completeness of data collection and to avoid missing important literature.

The reason for choosing these three databases for the literature search is that web of science Scopus and Google scholar basically cover research in social sciences, humanities, natural sciences, technology and education. We used the following keywords for the search: "Artificial Intelligence Education or AI Education" and "Educational Games, Gamification, Game-Based Learning". Finally, a total of 3348 pieces of literature were obtained. After eliminating the non-requested literature, the snowballing method was used to add the missing literature by scrolling forward and backward respectively. Snowballing is a method of obtaining additional literature by using references and citations from existing literature and providing additional information to a meta-

analysis dataset[14]. When using citations from the original article is referred to as forward snowballing and searching using references is referred to as reverse snowballing.

B. Inclusion Criteria

During the first phase of literature collection, we found that many of the papers were mostly related to the adoption of AI techniques to enhance learning, rather than teaching students how to learn about AI. There were also some papers that did not conduct an empirical study, but only a qualitative study, which did not meet the requirements of the papers needed for the meta-analysis. In order to obtain higher quality papers, we used the following literature inclusion criteria after eliminating duplicates:

The research entails the use of educational games or applications of relevant game elements for teaching and learning activities;

- Research topic is the effect of games on AI education
- The conclusion of the study needs to clearly report the final learning outcome indicators (learning achievement, engagement, work evaluation, etc.);
- The type of study is experimental or quasi-experimental, with pre- and post-test data reported for single-group experiments, and experimental and control groups for two-group experiments
- The study provides data that allow for complete calculation of effect sizes, such as sample size (N), mean (M), standard deviation (SD), etc.

After screening and compilation, a total of 15 initial samples of literature were obtained that met the requirements. We conducted a snowball search using these 15 studies as seed literature. The snowballing process yielded a total of 22 pieces of literature after three rounds of search, 5 in the first round, 2 in the second round, and 0 in the third round. In the end, a total sample of 22 eligible studies was obtained to include 46 analyzed effect values.

C. Coding Strategy

In order to facilitate later analysis and statistics as well as calculation of effect values, coding of eigenvalues of the sample studies was required before the implementation of the meta-analysis. To ensure the accuracy of coding, this study was conducted by two coders who independently coded 46 studies, and the coding consistency reached 98% with high confidence in the results. The coding of this study consisted of two main aspects.

First, the basic information of the literature, including the author's name (first author), date of publication, mean, standard deviation, t-value and p-value of the results of the experimental and control groups.

Second, the moderating variables were coded for seven main variables: learning effectiveness, academic period, experimental period, experimental size, knowledge type, game type and teacher intervention.

In our study, learning outcomes were classified into knowledge achievement, motivation and thinking skills. From

the included studies, the type of experimental outcomes that the studies ultimately tested varied widely, from studies that tested whether educational games were more conducive to improving students' AI knowledge achievement, to studies that tested whether educational games could promote students' motivation and thinking skills. The specific coding is shown in Table I.

TABLE I. META-ANALYSIS MODERATOR VARIABLE CODING TABLE

categories	Codes
Learning outcome	a1: Knowledge achievement a2: Motivation a3: Thinking skills
School segment	b1: Higher education b2: K-12
Experimental period	c1: Less than one week c2: one week to one month c3: more than one month
Experimental scale	d1: Small (0~40) d2: Medium (40~80) d3: Large (more than 80)
Knowledge type	e1: AI Concepts e2: AI Algorithms e3: Typical Applications
Game type	f1: Puzzle games f2: Role-playing games f3: Reasoning strategy games f4: Robot games
Teacher intervention	g1: Yes g2: No

D. Statistical Methods

Once the dataset has been collected, we will use CMA 3.3 software for meta-analysis. Effect size (ES) is a measure of the strength of an experimental effect, or the strength of the association of a variable, which is less affected by sample size and directly indicates the magnitude of the result, and can effectively help researchers understand whether a statistically significant difference is actually meaningful. In meta-analysis, Cohen's *d* and Hedges' *s g* values are often the most effective. For studies with large samples, there is little difference between the two effect sizes, whereas Cohen's *d* can grossly overestimate effect sizes in studies with small samples, and Hedges et al. added a correction factor to the *d* values to make Hedges' *s d* more accurate in studies with small samples[15]. In our study, the sample size and number of studies were small, so Hedges' *g* (referred to as *g* value) was used as the final effect size. And Cohen's proposed criteria for judging effect size were used: if $ES < 0.2$, it can be assumed that the intensity of the effect is very small; if $0.2 < ES < 0.5$, it can be assumed that there is a medium intensity of the effect; if $0.5 < ES < 0.8$, it can be assumed that there is an upper-medium intensity of the effect; and if $ES > 0.8$, it can be assumed that there is a large intensity of the effect [16].

Our study used a random-effects model. Differences in effect sizes between studies in a random-effects model are due not only to sampling error but also to systematic variance between studies[17]. As the 46 studies were conducted by independent researchers who implemented the interventions in different ways, we cannot assume that the heterogeneity between studies may not be affected by these different circumstances. In addition, the subsequent meta-analysis used different variables and study designs. Therefore, a random-effects model would have been more appropriate than a fixed effects model.

IV. RESULTS

A. Publication Bias

To ensure the scientific validity and reliability of research results, the meta-analysis process usually detects the bias of the sample data. In our study, we chose the funnel plot, Egger's test, Classic Fail-safe *N* and the trim and fill method to comprehensively assess the publication bias of the study sample. The funnel plot of the sample publication bias detection was drawn using CMA3.3 software, as shown in Figure 1. It can be seen that most of the studies are symmetrically distributed on both sides of the effect size and are located in the middle and upper part of the funnel plot. However, there are also studies that are not symmetrically and evenly distributed, indicating that publication bias may be present. As the results of the funnel plot may lead to subjective interpretation, we next analyzed them using Egger's test. Previous studies have mentioned that in Egger's test there is bias when $t > 1.96$, $p < 0.05$ and no bias when $t < 1.96$, $p > 0.05$ [18].

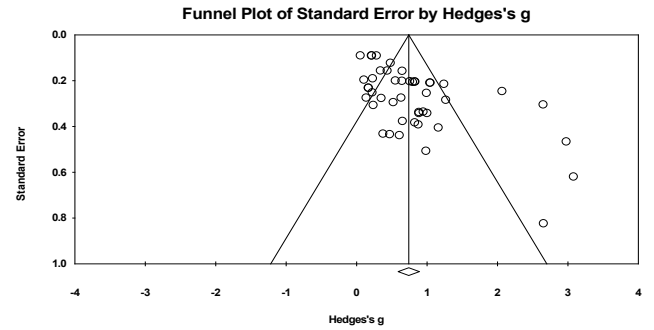


Fig. 1. Funnel plot of effect sizes with 95% confidence interval boundaries

Egger's test showed $t = 5.33$, $p < 0.05$, indicating possible publication bias. Next, the study used a classical fail-safe *N* to assess how many unpublished studies would be needed to bring the overall value of published studies to a non-significant level[19]. The measure of Classic Fail-safe *N* is $5 \cdot N + 10$, where *N* represents the number of literatures included in the meta-analysis in this study, so the measure for this study is $(5 \cdot 22 + 10) = 120$. After calculation by the CMA software, the coefficient of fail-safe in this study is 4698, which is much higher than 120. This metric suggests that the effect size of unpublished studies does not have much impact on the overall effect value of published studies. The overall effect size has little influence. Finally, we used a trim and fill method to assess the effect of publication bias on the robustness of the results under a random-effects model without the inclusion of dummy literature. The before-and-after results were unchanged, suggesting that there was no significant effect of publication bias on the magnitude of the combined effect.

TABLE II. RELIABILITY TESTS RESULTS REPRESENTING THE PROBABILITY OF PUBLICATION BIAS

Egger' s test	Classic Fail-safe N	Trim and Fill
t-value P	p-value>alpha	Observed/Added Trimmed
5.28 0.0	4628	0.74/0.74 0

B. Heterogeneity

All studies included in the same meta-analysis will inevitably differ from each other, and previous researchers have referred to the different variations between studies in a meta-analysis as heterogeneity[20]. Q-value and I² are commonly used in meta-analyses to test for heterogeneity. The criterion for the Q-test is usually set at 0.10. Heterogeneity exists between studies if $p < 0.10$.

Table 3 shows the combined effect values of the 46 studies and the results of the sample heterogeneity test showed that $Q=266.025$, $p=0.000 < 0.10$ and $I^2=83.084$. The results of the study showed that there was a large amount of heterogeneity between the samples, so the random effects model was chosen for the analysis.

TABLE III. SUMMARY OF THE EFFECT SIZES OBTAINED IN FIXED AND RANDOM-EFFECT META-ANALYSES

Model	ES(g)	95%CI		Heterogeneity		
		Lower	Upper	Q	P	I ²
Fixed	0.496	0.439	0.552	266.025	0.000	83.084
Random	0.740	0.589	0.891			

Note. K=the number of studies; ES=Effect size(g-value); 95%=95% confidence interval

C. Overall Effect of Educational Games in AI Education

The collection of 22 studies reported 46 independent effect sizes from measuring multiple outcomes. Some of the literature examined multiple variables related to learning effectiveness and therefore included more than one effect size. Based on the results of the heterogeneity test in the previous section, a random effects model was selected for the analyses, with a combined effect size of 0.74. As mentioned above in the criteria for classifying effect sizes, the effect sizes for this study ranged from $0.5 < ES < 0.8$. This shows that educational games have a moderately positive effect on AI learning and can improve student learning more significantly.

D. Learning Outcome

In our meta-analysis study, the learning outcomes of AI were measured in different ways, categorized as: knowledge achievement, motivation and thinking skills. The results in Table 4 show no heterogeneity between groups ($Q=10.853$, $p=0.004 < 0.05$). Twenty-four studies showed that educational games were more conducive to improving students' knowledge achievement ($g=0.841$, $p < 0.05$). Fifteen studies tested students' motivation and showed that educational games were more conducive to improving students' motivation ($g=0.743$, $p < 0.05$). The next seven studies reported on the development of students' thinking skills ($g=0.367$, $p < 0.05$) and found that educational games had a moderately low effect on the development of thinking skills.

E. Moderator Analysis

To test whether the moderating variables had an influential effect on students' AI knowledge. Table 5 summarizes the results of the analyses of the moderating variables under the random effects model, including academic year, experimental period, size of experiment, type of AI knowledge, type of game and teacher intervention.

In this study, the experimental period was divided into three levels, less than one week ($n=13$), between one week and one month ($n=12$) and more than one month ($n=16$), all with g values between 0.6 and 0.7, with possible heterogeneity between groups ($QB=2.584$, $p > 0.05$). The largest effect sizes were found for experimental periods of more than one month ($g=0.690$, $p < 0.05$) and the smallest for less than one week ($g=0.654$, $p < 0.05$).

The experimental scale was classified as small with less than 40 people, medium with between 40 and 80 people and large with more than 80 people. The results showed that there was no heterogeneity between the experimental size groups ($QB=6.510$, $p < 0.05$), the highest effect was found in the small-scale ($g=1.119$, $p < 0.005$), the medium-scale effect was $g=0.661$ ($p < 0.05$), and the smallest was found in the large-scale ($g=0.608$, $p < 0.05$), and the effect values of all three exceeded 0.6.

The distribution of the number of studies showed that educational games were used in the largest number of studies for teaching AI algorithms ($n=19$) and AI concepts ($n=18$), with a smaller number of typical applications ($n=9$). There was less heterogeneity between groups ($QB=6.523$, $p < 0.05$), for teaching AI concepts the effect size was $g=0.524$ ($p < 0.05$), for AI algorithms $g=0.957$ ($p < 0.05$) and for typical applications $g=0.797$ ($p < 0.05$), and all were statistically significant.

There was less heterogeneity between groups for game types ($QB=7.825$, $p=0.05$). Puzzle games had the largest effect size ($g=0.948$, $p < 0.05$), while reasoning strategy games had the smallest effect size ($g=0.513$, $p < 0.05$). Robotics games had the least use ($n=5$) with an effect size of $g=0.826$ ($p < 0.05$). Role-playing and puzzle games were used the most, with an effect size of $g=0.741$ ($p < 0.05$) for role-playing games.

In our study, we also analyzed the academic segments and found that educational games were used in higher education AI courses in higher numbers ($n=31$) and with higher effect sizes ($g=0.867$, $p < 0.05$). In contrast, educational games were used in K-12 AI courses in smaller numbers ($n=14$) and with smaller effect sizes ($g=0.414$, $p < 0.05$). Some of the AI courses in the included studies were found to introduce educational games with teacher intervention ($g=0.751$, $p < 0.05$), while others had no teacher intervention ($g=0.651$, $p < 0.05$), allowing students to conduct their own workshops.

V. DISCUSSION

For Q1, the results of the study showed that educational games have a moderately high level of positive impact on AI education ($g = 0.740$). This is in line with previous qualitative studies analyzing the use of educational games in AI education, which also concluded that educational games help to teach AI knowledge.

To understand the impact of educational games specifically on the dimensions of students' AI learning outcomes, this study analyzed three dimensions: knowledge achievement, motivation to learn and thinking skills. It was found that educational games had the greatest impact on knowledge achievement ($g=0.841$) and motivation to learn ($g=0.743$), while thinking skills had a smaller impact ($g=0.367$). This is also more in line with previous research, which shows that educational games create a pleasant and positive learning atmosphere that can effectively stimulate

interest in learning [21]. It can also be argued that educational games can be used as a teaching tool to effectively teach AI-related knowledge and improve learning outcomes. For the development of thinking skills, the effect value of educational games is lower, which may be due to the following reasons: the teaching process of gamification is more oriented towards the process of entertainment, whereas the development of thinking skills requires a logical and rigorous teaching process to guide it.

The moderated effects results showed that educational games had a greater impact on higher education ($g=0.867$) than

on K-12 education ($g=0.414$). This is a departure from what we had previously expected, and there are possible reasons for this. First, AI is more widespread and deeper in higher education, and consequently the use of educational games is more diverse in higher education, which may have a greater positive impact on students' knowledge achievement and motivation. In K-12 education, AI is more obscure and difficult for students at this stage. And learning content such as algorithms and programming imposes a greater cognitive load on students at this stage. Educational games can be beneficial for K-12 students in terms of stimulating their interest and motivation in learning AI, but not be as beneficial for content learning.

TABLE IV. LEARNING OUTCOMES ANALYSIS

Learning outcome	n	ES	Standard error	Random-effects model 95%CI		Test of null(2-Tail)		Heterogeneity
				Lower	Upper	Z-value	P-value	
Knowledge	24	0.841	0.129	0.588	1.095	6.510	0.00	Q=10.853 df(Q)=2 p=0.004
Motivation	15	0.743	0.126	0.497	0.989	5.911	0.00	
Thinking skills	7	0.367	0.094	0.183	0.552	3.899	0.00	

Note: n=the number of studies; ES=effect size (g value)

TABLE V. RESULT OF MODERATOR ANALYSIS

Moderator Variables	n	Random-effects model		95%CI		QB	P
		Effect size		Lower	Upper		
		g	p				
School segment						12.285	0.002
Higher education	31	0.867	0.000	0.665	1.069		
K-12	14	0.414	0.000	0.243	0.585		
Experimental period						2.584	0.460
Less than one week	13	0.654	0.000	0.337	0.970		
One week to one month	12	0.661	0.000	0.388	0.934		
More than one month	16	0.690	0.000	0.536	0.844		
Experimental scale						6.510	0.039
0~40(small)	15	1.119	0.000	0.765	1.473		
40~80（medium）	11	0.661	0.002	0.248	1.073		
More than 80（large）	20	0.608	0.000	0.435	0.780		
Knowledge type						6.523	0.038
AI Concepts	18	0.524	0.000	0.370	0.678		
AI Algorithms	19	0.957	0.000	0.635	1.279		
Typical Applications	9	0.797	0.000	0.417	1.177		
Game type						7.825	0.050
Puzzle games	14	0.948	0.000	0.625	1.271		
Role-playing games	14	0.741	0.000	0.552	0.931		
Reasoning strategy games	13	0.513	0.000	0.367	0.660		
Robot games	5	0.826	0.008	0.213	1.439		
Teacher intervention						0.238	0.626
Yes	39	0.751	0.000	0.588	0.915		
No	7	0.651	0.001	0.281	1.021		

Note: n=the number of studies; SE=standard error; QB=Q statistic testing the between-group variation; P=Significance value of the test for subgroup difference

For Q4, the results of the study show that the educational game becomes more effective for learning about AI the longer the experimental period. The effect values for experimental cycles of less than a week and more than a month are both around 0.6, and the effect of the length of the experimental cycle is present but not obvious. This may be due to that in the early stages of adaptation, students gradually lose interest in the novelty of the game itself and eventually return to the knowledge content itself.

The results of the above study show that the use of educational games to teach AI knowledge is best in small classes ($g=1.119$). This is also consistent with previous research findings that teachers in small classes are better able to pay

attention to each student's learning outcomes and provide the necessary help in a timely manner[22]. If the class size is too large, it may not be possible for students to know and understand how to use or play the game in time. It is also not possible to complete all the tasks and objectives of the game assigned by the teacher in time.

Considering the content composition of AI knowledge, to explore the effect of educational games on different types of AI knowledge, we categorized AI knowledge into AI concepts, AI algorithms and typical applications. The results of the study show that educational games are more effective in teaching AI algorithms ($g = 0.957$) and typical applications ($g = 0.797$), and less effective in teaching AI concepts. The reason may be that knowledge of AI algorithms and applications is biased towards

practical operation, and educational games teach knowledge to students in the form of a game, and the process of the game is also the process of practice, requiring students to manipulate the game character to complete the task. Therefore, educational games and AI algorithms and applications can be well combined, so that students can acquire knowledge in the process of practical practice. As for AI concepts, the type of knowledge belongs to declarative knowledge, students may be too involved in the process of the game in the entertainment of the game, for some conceptual and textual knowledge stays in the short-term memory, so the learning effect will be smaller.

Considering the effects of the Q7 study, we categorized the types of games (i.e. puzzle games, role-playing games, reasoning and strategy games, and robotics games). The results showed that all four types of games included in this study had a positive impact on students' learning of AI, with puzzle games having the greatest positive impact. Puzzle games are more likely to combine gameplay with content, and the form of the game is simpler, which is easier for students to accept, so they can gain knowledge by completing levels and solving puzzles. The impact of robotics games is also higher, probably because the form of robotics games is closer to the type of structure of AI knowledge, and students learn about the construction and principles of the robot as they design and build it. Role-playing and strategy games, on the other hand, have a slightly lower impact effect than the former, probably because these two types of games imply a more complex storyline, which does not fit well with the practical type of AI knowledge.

To explain Q8, we finally added a moderating variable, i.e. having the wrong teacher intervention. The results of the study showed that the number of studies with teacher intervention in the gamified learning process was higher and the final educational outcomes were better, which is also in line with the previously expected results. Educational games are engaging for students, but it can be difficult for novice students to learn the content of the game in a limited amount of time. Therefore, there is also a need for teacher intervention to help students get up to speed more quickly and to reduce their cognitive load. The study without teacher intervention also had a medium impact effect, probably because the absence of teacher intervention is more likely to stimulate students' ability to think independently and better develop their thinking skills, but it is also more demanding for students.

VI. CONCLUSION

This study verified the positive impact of educational games on AI education through a meta-analysis approach. Specifically, educational games have the greatest impact on knowledge achievement and motivation to learn, followed by thinking skills. The possible reason for this is that the process of entertainment in the game helps to reduce the cognitive load of the learning process and is conducive to stimulating students' interest in learning. With the current rapid development of metaverse and generative AI technologies, studies have already begun to apply such technologies to enhance the process of gamification in the curriculum. We will pay more attention to this type of research in the future, and more of this type of research needs to be included in future reviews.

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